

Control Structures Observation

Tables used: STUDENT, FACULTY, DEPT, COURSE, ENROLLMENT.

AIM: To use control structures — IF, CASE, WHILE, REPEAT, and LOOP — inside MySQL stored programs.

Control structures only work inside a BEGIN...END block (procedure, function, or trigger). DELIMITER must be changed while defining them.

-- 1. IF / ELSEIF / ELSE

```
DELIMITER //
CREATE PROCEDURE GradeFaculty(IN age INT)
BEGIN
    IF age >= 60 THEN
        SELECT 'Senior' AS Grade;
    ELSEIF age >= 45 THEN
        SELECT 'Mid-level' AS Grade;
    ELSE
        SELECT 'Junior' AS Grade;
    END IF;
END //
DELIMITER ;

CALL GradeFaculty(55);
```

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE GradeFaculty(IN age INT)
-> BEGIN
->     IF age >= 60 THEN
->         SELECT 'Senior' AS Grade;
->     ELSEIF age >= 45 THEN
->         SELECT 'Mid-level' AS Grade;
->     ELSE
->         SELECT 'Junior' AS Grade;
->     END IF;
-> END //
Query OK, 0 rows affected (0.04 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL GradeFaculty(55);
+-----+
| Grade |
+-----+
| Mid-level |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

-- 2. CASE statement (simple form)

```
DELIMITER //
CREATE PROCEDURE DeptFullName(IN code VARCHAR(10))
BEGIN
    CASE code
        WHEN 'CSE' THEN SELECT 'Computer Science Engineering' AS FullName;
        WHEN 'ECE' THEN SELECT 'Electronics and Communication' AS FullName;
        WHEN 'IT' THEN SELECT 'Information Technology' AS FullName;
        ELSE SELECT 'Unknown Department' AS FullName;
    END CASE;
END //
DELIMITER ;

CALL DeptFullName('CSE');
```

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE DeptFullName(IN code VARCHAR(10))
-> BEGIN
->     CASE code
->         WHEN 'CSE' THEN SELECT 'Computer Science Engineering' AS FullName;
->         WHEN 'ECE' THEN SELECT 'Electronics and Communication' AS FullName;
->         WHEN 'IT' THEN SELECT 'Information Technology' AS FullName;
->         ELSE SELECT 'Unknown Department' AS FullName;
->     END CASE;
-> END //
Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL DeptFullName('CSE');
+-----+
| FullName |
+-----+
| Computer Science Engineering |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

-- 3. CASE statement (searched form, with conditions)

```
DELIMITER //
CREATE PROCEDURE AgeCategory(IN age INT)
BEGIN
    CASE
        WHEN age < 40 THEN SELECT 'Young' AS Category;
        WHEN age BETWEEN 40 AND 55 THEN SELECT 'Middle-aged' AS Category;
        ELSE SELECT 'Senior' AS Category;
    END CASE;
END //
DELIMITER ;
```

```
CALL AgeCategory(50);
```

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE AgeCategory(IN age INT)
-> BEGIN
->     CASE
->         WHEN age < 40 THEN SELECT 'Young' AS Category;
->         WHEN age BETWEEN 40 AND 55 THEN SELECT 'Middle-aged' AS Category;
->         ELSE SELECT 'Senior' AS Category;
->     END CASE;
-> END //
Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL AgeCategory(50);
+-----+
| Category |
+-----+
| Middle-aged |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

-- 4. WHILE loop

```
DELIMITER //
CREATE PROCEDURE CountUpTo(IN limitVal INT)
BEGIN
    DECLARE i INT DEFAULT 1;
    WHILE i <= limitVal DO
        SELECT i AS CurrentValue;
        SET i = i + 1;
    END WHILE;
END //
DELIMITER ;

CALL CountUpTo(3);
```

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE CountUpTo(IN limitVal INT)
-> BEGIN
->     DECLARE i INT DEFAULT 1;
->     WHILE i <= limitVal DO
->         SELECT i AS CurrentValue;
->         SET i = i + 1;
->     END WHILE;
-> END //
```

Query OK, 0 rows affected (0.02 sec)

```
mysql> DELIMITER ;
mysql>
mysql> CALL CountUpTo(3);
```

```
+-----+
| CurrentValue |
+-----+
|             1 |
+-----+
1 row in set (0.00 sec)
```

```
+-----+
| CurrentValue |
+-----+
|             2 |
+-----+
1 row in set (0.00 sec)
```

```
+-----+
| CurrentValue |
+-----+
|             3 |
+-----+
```

-- 5. REPEAT ... UNTIL loop

```
DELIMITER //
CREATE PROCEDURE CountDownFrom(IN startVal INT)
BEGIN
    DECLARE i INT DEFAULT startVal;
    REPEAT
        SELECT i AS CurrentValue;
        SET i = i - 1;
    UNTIL i = 0
```

```
END REPEAT;  
END //  
DELIMITER ;  
  
CALL CountdownFrom(3);
```

```
mysql> DELIMITER //  
mysql> CREATE PROCEDURE CountdownFrom(IN startVal INT)  
-> BEGIN  
->     DECLARE i INT DEFAULT startVal;  
->     REPEAT  
->         SELECT i AS CurrentValue;  
->         SET i = i - 1;  
->     UNTIL i = 0  
->     END REPEAT;  
-> END //  
ERROR 1304 (42000): PROCEDURE CountdownFrom already exists  
mysql> DELIMITER ;  
mysql>  
mysql> CALL CountdownFrom(3);  
+-----+  
| CurrentValue |  
+-----+  
|          3 |  
+-----+  
1 row in set (0.00 sec)  
  
+-----+  
| CurrentValue |  
+-----+  
|          2 |  
+-----+  
1 row in set (0.00 sec)  
  
+-----+  
| CurrentValue |  
+-----+  
|          1 |  
+-----+  
1 row in set (0.01 sec)  
  
Query OK, 0 rows affected (0.01 sec)
```

-- 6. LOOP with LEAVE (like a break statement)

```
DELIMITER //  
CREATE PROCEDURE LoopWithBreak()  
BEGIN  
    DECLARE i INT DEFAULT 1;
```

```

myloop: LOOP
    IF i > 3 THEN
        LEAVE myloop;
    END IF;
    SELECT i AS Value;
    SET i = i + 1;
END LOOP myloop;
END //
DELIMITER ;

CALL LoopWithBreak();

```

```

mysql> CREATE PROCEDURE LoopWithBreak()
-> BEGIN
->     DECLARE i INT DEFAULT 1;
->     myloop: LOOP
->         IF i > 3 THEN
->             LEAVE myloop;
->         END IF;
->         SELECT i AS Value;
->         SET i = i + 1;
->     END LOOP myloop;
-> END //

```

Query OK, 0 rows affected (0.01 sec)

```

mysql> DELIMITER ;
mysql>
mysql> CALL LoopWithBreak();

```

```

+-----+
| Value |
+-----+
|      1 |
+-----+
1 row in set (0.00 sec)

```

```

+-----+
| Value |
+-----+
|      2 |
+-----+
1 row in set (0.00 sec)

```

```

+-----+
| Value |
+-----+
|      3 |
+-----+
1 row in set (0.00 sec)

```

Query OK, 0 rows affected (0.01 sec)

-- 7. ITERATE (like a continue statement)

```
DELIMITER //
CREATE PROCEDURE SkipEvenNumbers()
BEGIN
    DECLARE i INT DEFAULT 0;
    myloop: LOOP
        SET i = i + 1;
        IF i > 5 THEN
            LEAVE myloop;
        END IF;
        IF i % 2 = 0 THEN
            ITERATE myloop;
        END IF;
        SELECT i AS OddValue;
    END LOOP myloop;
END //
DELIMITER ;

CALL SkipEvenNumbers();
```

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE SkipEvenNumbers()
-> BEGIN
->     DECLARE i INT DEFAULT 0;
->     myloop: LOOP
->         SET i = i + 1;
->         IF i > 5 THEN
->             LEAVE myloop;
->         END IF;
->         IF i % 2 = 0 THEN
->             ITERATE myloop;
->         END IF;
->         SELECT i AS OddValue;
->     END LOOP myloop;
-> END //
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> DELIMITER ;
mysql>
mysql> CALL SkipEvenNumbers();
```

```
+-----+
| OddValue |
+-----+
|         1 |
+-----+
1 row in set (0.00 sec)
```

```
+-----+
| OddValue |
+-----+
|         3 |
+-----+
1 row in set (0.00 sec)
```

```
+-----+
| OddValue |
+-----+
|         5 |
+-----+
1 row in set (0.00 sec)
```

```
Query OK, 0 rows affected (0.00 sec)
```